

# YUANJUN (JOHNNY) DAI

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## SUMMARY

**AI Infrastructure and Applied AI Engineer, PhD candidate** at CWRU. Specializes in the security, optimization, and observability of ML/LLM training and inference infrastructure. 8 research papers (7 first-author) at IEEE, ACM, and Springer; **2+ years of systems software engineering** at Cisco and Broadcom; 2+ years of **cross-functional** product ownership at Midea and Xiaomi.

## EDUCATION

Ph.D. Candidate in Computer Science (AI Infrastructure) 08/2019 – Sep. 2026 (exp.)  
Case Western Reserve University

M.S. in Electrical and Computer Engineering (Computer Architecture)  
University of Pittsburgh

B.S. in Electrical Engineering  
Northeastern University

## SKILLS

|                       |                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------|
| Languages             | Python, C/C++                                                                           |
| ML Frameworks         | PyTorch (DDP/FSDP), vLLM, Megatron-LM, NCCL, BytePS, scikit-learn                       |
| Parallelism Paradigms | Data Parallel (DDP/FSDP), Tensor Parallel (TP), Pipeline Parallel (PP), Hybrid Parallel |
| Infrastructure        | Kubernetes, Docker, CI/CD Pipelines (GitHub Actions), ArgoCD, Slurm                     |
| Systems & APIs        | FastAPI, REST API, gRPC                                                                 |
| Observability         | eBPF (Kernel Tracing), Prometheus, Grafana, DCGM Exporter, Intel RAPL                   |
| Cloud & HPC           | Lambda Labs HPC, Ohio Supercomputer Center, NSF FABRIC, CloudLab                        |
| Networking            | SDN (Software-Defined Networking), P4, NPU/ASIC (Broadcom BCM SDK)                      |

## AI INFRASTRUCTURE & SYSTEMS EXPERIENCE

**Research Assistant, AI Infrastructure & ML Systems** 08/2019 – Sep. 2026 (exp.)  
Case Western Reserve University

Research focus: **security, optimization, and observability** of **ML/LLM** training and inference infrastructure.

### DYNAMIX – Reinforcement Learning-Based Adaptive Batch Scheduling for Distributed ML | [Paper](#) / [Code](#)

- Designed a **PPO**-based **Reinforcement Learning (RL)** scheduler monitoring hardware signals (CPU/memory, bandwidth, retransmission rate via **eBPF** probes) and statistical signals (loss convergence, generalization) to dynamically adapt batch size
- Co-scales learning rate via **Linear Scaling Rule**; evaluated on **PyTorch DDP** and **Parameter Server (BytePS)**
- Achieves **46% wall-clock speedup** and **~6% accuracy gain** across **8–64 A100 GPUs** over fixed batch size baseline

### PRISM – Priority-Based Selective Reliability for Distributed ML Training | [Paper](#)

- Designed a transport-layer library (**NCCL/Gloo** → **UDP**) hooking into **PyTorch DDP**'s AllReduce at bucket-ready events; classifies gradients by magnitude into high-priority (full **SACK**-based retransmission) and low-priority (local compensation via cached values)
- Supports kernel-bypass (**LibVMA**) and kernel-based (**GSO/GRO**) variants; achieves **8–15% throughput gain** for **CNNs** and **15–21% for LLMs** over TCP, matching NDP latency at 144-worker scale

### Scalable LLM Serving Infrastructure on Kubernetes | [Code](#) / [Video](#)

- Independently designed a self-hosted **K8s** LLM inference platform (NGINX → **FastAPI** → **vLLM**) with **GitOps CI/CD** (**ArgoCD** + GitHub Actions) across 50+ deployments; validated on **RTX 4050** (**GPU time-slicing**, **Horizontal Pod Autoscaler (HPA)** 1↔2) and **Lambda Labs A100** (**Multi-Instance GPU (MIG)** 7×1g.5gb, **HPA** 1↔7) with **91.3% peak GPU compute** under extreme stress
- Built 4-layer GPU observability (**DCGM Exporter** + **Prometheus** + **Grafana**: GPU hardware, vLLM metrics, FastAPI gateway, K8s cluster)

### Domain-Adapted LLMs for Clinical Risk Adjustment Coding | [Paper](#)

- Built a 115-label HCC multi-label classifier on highly imbalanced EHR data (MIMIC-V), improving long-tail performance via **inverse-frequency weighted binary cross-entropy loss** and **per-label threshold calibration**, achieving **0.775 macro-F1 / 0.742 micro-F1 / 0.9362 macro-AUC**.
- Trained **PubMedBERT** using **PyTorch DDP** on 2×NVIDIA RTX A6000 GPUs for efficient scaling, while fine-tuning **Qwen-7B** using distribution-aware **few-shot prompting** (5-shot with negative, frequent, and rare-label demonstrations) to **mitigate hallucination** and enforce structured HCC outputs with improved sensitivity to sparse labels; scaled full-parameter training via **NeMo** + **Megatron-LM** (TP=1, PP=1) with **PyTorch FSDP (ZeRO-3)**, **FlashAttention**, and **activation checkpointing**, enabling training under GPU memory constraints (avoiding OOM).

## AI-Powered Job Hunting Agent | [Code](#)

- Built a production-oriented **multi-agent AI platform** for automated job discovery, semantic matching, and personalized resume optimization across LinkedIn, Indeed, Dice, YC Jobs, and HackerNews
- Designed a **LangGraph-based orchestration pipeline** with specialized agents for job collection, matching, resume refinement, trend analysis, and workflow validation
- Developed a **6-dimension LLM-based job matching framework** via careful **prompt engineering** to evaluate skills, experience, seniority, work authorization, and company preference beyond keyword filtering
- Implemented a **capability-aware multi-model routing pipeline** for analytical planning and recruiter-oriented resume optimization
- Engineered scalable backend infrastructure with **FastAPI**, cross-platform job deduplication, health monitoring endpoints, and deployment-ready automation services

## Priority-Based eBPF Network Flow Telemetry for AI / DML Infrastructure | [Paper](#) / [Code](#)

- Designed PSketch, the first in-kernel priority-aware sketch using **eBPF**; high-priority flows tracked via in-kernel hashmap ( $O(1)$ ), low-priority via **3-layer Count-Min Sketch**; enables  **$O(n)$  Top-K detection** with sub-linear memory, voting-based collision resolution and atomic operations for thread-safe updates
- Achieves **96% Top-K accuracy**, **96.4% retransmission recall**, and  $<1.1\%$  throughput overhead at 10 Gbps on FABRIC testbed

## High-Resolution eBPF System Resource Profiling for DML Workloads | [Paper](#) / [Code](#)

- Designed eHashPipe, a dual-pipeline **eBPF** framework with  **$O(n)$  Top-K tracking**: (1) HashPipe-based memory profiler with cascading slot eviction and ptr2size deallocation tracking, (2) per-thread CPU tracker via sched\_switch tracepoint
- Delivers **10–14× finer temporal resolution** than top/htop,  $\sim 11\text{ms}$  latency, and  $>90\%$  Top-K fidelity on commodity Linux servers

## Training-Time Side-Channel Attacks on Distributed ML Systems | [Paper](#) / [Code](#)

- **Training-phase black-box attack** on **MXNet BSP** distributed training, reconstructing full DNN architectures via two unprivileged side channels: **Intel RAPL** (PP0+DRAM,  $50\mu\text{s}$  via powercap/sysfs) and 0-payload network packet sequences (**Seq\_T**) for tensor size extraction
- **Lookahead** (sliding-window SMA) + **Lookbehind** algorithms isolate forward passes; **band-pass filtering** segments individual layers; **Decision Tree** (entropy, 97.7%, 10-fold CV on 84K+ segments) and **BiLSTM** classify layer types; weighted-average **MLP ensemble** (weights  $\propto 1/\text{validation error}$ ) recovers Conv/FC dimensions
- Achieves  **$>97\%$  layer-type accuracy**; validated across 20 DNN models (LeNet through VGG/ResNet families)

## INDUSTRY EXPERIENCE

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### Software Engineer II

06/2015 – 08/2016

Cisco Systems, Inc.

- Developed NPU/ASIC features for a 100 Gbps core router (Azure, AWS, Alibaba Cloud) using Broadcom BCM SDK.
- Contributed to QoS and ECMP modules, reducing forwarding latency under heavy load.

### Software Engineer Intern

10/2014 – 05/2015

Broadcom (Emulex)

- Re-architected RTOS data acquisition with DMA + circular-buffer, **reducing CPU load by 30%**; ported BladeEngine ASIC management CLI from Linux to Windows Server.

### Product Manager

09/2016 – 06/2018

Midea Group

- Led technical PM for **Fun Oven II** across ML model training, Docker deployment on Alibaba Cloud, mobile app, and culinary algorithm design; won **Red Dot Design Award**, debuted at 2018 AWE Show · [Red Dot ↗](#) · [PR Newswire ↗](#)

### Technical Product Manager

07/2018 – 01/2019

Xiaomi

- Defined IoT appliance product requirements via competitive analysis and user research; assessed technical feasibility and delivered architecture proposals to align engineering and business stakeholders.

## PUBLICATIONS

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8 papers total, 7 first-author, 5 published at IEEE CCNC, IEEE ICNC, ACM TOIT, and Springer SecureComm; 3 under review (incl. ACM TOPS). Full list: [johnny-dai-git.github.io/portfolio](https://johnny-dai-git.github.io/portfolio)

## PATENTS

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Oven Temperature Control Method, Device and Computer-Readable Storage Medium — Granted CN 108594898 B | Issued July 23, 2021 | Assignee: Midea Group Co., Ltd. | Dai Yuanjun et al.